Veronica Kelsey

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Work and Education

2021 - 2027 **Heilbronn Research Fellow** The University of Manchester

2018 - 2021 **PhD Mathematics** University of St Andrews
Supervisors: Prof Colva M. Roney-Dougal and Dr Martyn Quick
Thesis: Base size and generating graphs of primitive permutation groups

2014 - 2018 **MMath Mathematics** The University of Manchester
Dissertation: Mathieu groups and Chamber Graphs

Peer Reviewed Publications

[1] **The excess zero graph of a Coxeter group**
with S.B. Hart and P.J. Rowley - J. Group Theory (To appear)

[2] **Root cycles in Coxeter groups**
with S.B. Hart and P.J. Rowley - J. Group Theory (2025)

[3] **The relational complexity of linear groups acting on subspaces**
with S.D. Freedman and C.M. Roney Dougal - J. Group Theory (2025)

[4] **A note on the rank 5 polytopes of $\mathbf{M_{24}}$**
with R. Nicolaides and P.J. Rowley - Innov. Incidence Geom. (2024)

[5] **On the rank 5 polytopes of the Higman-Sims simple group**
with R. Nicolaides and P.J. Rowley - Innov. Incidence Geom. (2022)

[6] **On relational complexity and base size of finite primitive groups**
with C.M. Roney-Dougal - Pacific J. Math. (2022)

[7] **Maximal cocliques in the generating graphs of alternating and symmetric groups**
with C.M. Roney-Dougal - Comb. Theory (2022)

[8] **Generating finite Coxeter groups with elements of the same order**
with S.B. Hart and P.J. Rowley - Turkish J. Math. (2021)

[9] **Chamber graphs of minimal parabolic sporadic geometries**
with P.J. Rowley - Innov Incidence Geom. (2020)

[10] **A note on involution centralizers in black box groups**
with P.J. Rowley - J. Group Theory (2020)

[11] **Chamber graphs of some geometries that are almost buildings**
with P.J. Rowley - Innov. Incidence Geom. (2019)

[12] **$\mathbf{M_{24}}$ orbits of octad triples**
with P.J. Rowley - Graphs Combin. (2018)

Preprints and Papers in Preparation

[1] **$\mathbb{P}(q)$-Groupoids of Conway Type**
with P.J. Rowley - submitted - arXiv:2410.22202

[2] **Matroids, IBIS groups and Steiner systems**
with C.F. Nyberg-Brodda - in preparation

[3] **Minimal relations of relational structures**
with S.D. Freedman and C.M. Roney Dougal - in preparation

[4] **Imprimitive maximal cocliques in the generating graphs of $\mathbf{A}_n$ and $\mathbf{S}_n$**
with C.M. Roney-Dougal - in preparation

[5] **On String C-Group Representations of Covers of the Sporadic Simple Groups**
with P.J. Rowley - in preparation

Invited Visits

2025 **Korea Institute for Advanced Study** Visiting scholar and speaker, 2 weeks
2023 **CMI-HIMR Summer School in Probabilistic Group Theory** Teaching assistant and speaker
2022 **Young Group theorists workshop** Participant and speaker at SwissMap Research Station
2022 **MFO Research Visitor** Research collaboration in Oberwolfach, 2 weeks
2022 **INI Research Visitor** Research visit and workshop attendee, 2 weeks
2020 **Research in "Pairs"** Research collaboration at Oberwolfach, 2 weeks
2020 **INI Research Visitor** Research visit and workshop attendee, 2 weeks

Awards and Funding

2023-25 **HIMR small grant £1840 p.a.** Funding for a research seminar
2024 **STEM for Britain finalist** Poster competition and exhibition held in Parliament
2023 **HIMR small research grant £1000** Funding to host a collaborator for 1 week
2023 **MIMS funding £700** Funding to host a collaborator for 1 week
2018 **St Andrews PhD scholarship** Fees and stipend for 3 years
2018 **Outstanding Academic Achievement Award £1000** Top 0.5% of Manchester students
2018 **IMA Prize 2018** Final year outstanding project award
2018 **Manchester President's Doctoral Scholar Award** Fees and stipend for 3.5 years (declined)
2018 **Nominated for The Distinguished Achievement Award** Undergraduate of the Year
2017 **Fourth Year Scholarship** Tuition fee waiver
2017 **LMS Undergraduate Research Bursary £1440** Funding for a summer research project
2016 **BP achievement award £2000** For an essay on Big Data
2015-6 **Dalton scholarship £300** Academic achievement in 1st and 2nd year

Invited Talks

Jul 2026 **Groups arising from puzzles and games** Postgraduate Group Theory Conference
Feb 2026 **Matroids and IBIS groups** Bristol Algebra Seminar
Oct 2025 **Irredundant bases for permutation groups** Korean Institute of Advanced Study, Combinatorics And Representation Theory Seminar
Sep 2024 **Base and irredundant base size - how different are they really?** Topics in Group Theory - on the occasion of Lucchini's 60th birthday, Padova
Sep 2024 **Cliques and cocliques in generating graphs** Dualities in Prob. and Alg., Lancaster
Jun 2024 **My favourite group** British Early Career Mathematicians' Colloquium, Birmingham
Nov 2023 **Generating groups** Birmingham Algebra Seminar
Oct 2023 **Relational complexity of linear groups** Warwick Algebra Seminar
Mar 2023 **Calculating relational complexity** London Algebra Colloquium
Sep 2022 **Generating graphs and feeling good enough** Piscopia Society PiWorks seminar
Sep 2022 **A survey of base size** Young Group Theorists Workshop
Feb 2022 **Relational complexity** St Andrews Pure Colloquium
Nov 2021 **Group statistics** Bristol Junior Algebra Colloquium
Nov 2021 **Maximal irredundant base size of linear groups** Birmingham Algebra Seminar
Jan 2021 **Relational complexity of permutation groups** Functor Categories meeting
Oct 2020 **Base size of permutation groups** North British Semigroups Network
Jun 2020 **M_{24} and octad triples** Junior London Algebra Colloquium
May 2020 **Maximal subgroups and maximal cocliques** York Algebra Seminar

Teaching

2025 **Galois Theory** Co-leading a level 4 course with 22 students
Personal responsibilities: 12 lectures, 6 tutorials, midterm and exam marking

2025 **Coding Theory** Wrote and delivered a 12 hour course for 10 postdocs

2023 **O'Nan Scott Theorem** Lead a reading group for 8 people

2015-21 **Tutorial lead** Various undergraduate modules at Manchester & St Andrews University
Personal responsibilities: lead tutorials and marked work for approx 15 students per class

Outreach, EDIA and Mentoring

2024-25 **Manchester EDIA Forum** Built a postdoc EDIA webpage with resources and created a poster for Black History Month

2024-25 **HIMR EDIA Working group** Volunteered at a pre interview event, helped create a culture survey, and collated advice on giving accessible talks

2023-25 **LMS Levelling Up** Ran weekly tutorials and created revision resources for five A-level students from under-represented backgrounds

2025 **Inclusivity in Mathematical sciences** Participated in two workshops focusing on creating resources which highlight the work of non-western mathematicians and promote EDIA

2025 **EDIA "reading" group** Founded a reading group focusing on more niche EDIA topics with a book, podcast and film for each session to ensure accessibility
Previous: Freedom House EMS, 504 sit-in. Upcoming: social housing design, Welsh Not

2018-23 **Mentor** Volunteered as a mentor in peer mentoring schemes for undergraduate and masters students at the universities of Manchester and St Andrews

2024 **Mathematical Modelling and Problem Solving** Lead a workshop on cryptography for local secondary school students

2024 **MAP supervisor** Created and supervised two projects for A-level students on the Manchester Access Programme, a widening participation scheme

2024 **Martingale Foundation** Academic interviewer for scholarships for individuals from low-socioeconomic backgrounds

2023 **Women in Maths Day** Created and ran a interactive workshop session on mazes for GCSE and A-level students

Other Activities

Coding Experienced in Magma and GAP
British Sign Language Obtained Level 2, studying for Level 3
Ultra running 50 mile and 50k races, started an instagram community with over 30k followers
Braille Learning level 1

References

Prof Alexandre Borovik MMath Dissertation Marker alexandre@borovik.net
Prof Sarah Hart Collaborator s.hart@bbk.ac.uk
Prof Colva M. Roney-Dougal PhD Supervisor colva.roney-dougal@st-andrews.ac.uk